TTK4190 Guidance, Navigation and Control of Marine Craft, Aircraft and Drones

Assignment 2 Part 1

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1 Problem 1

1.1 1a

The moment of inertia about the z-axis through the center of gravity is given by:

$$I_z^{CG} = \int_V \left(x^2 + y^2\right)\, dm \tag{1}$$

Assuming homogeneously mass distribution, we have $dm = \rho_m\, dV$, therefor:

$$I_z^{CG} = \rho_m \int_V \left(x^2 + y^2\right)\, dV \tag{2}$$

A rectangular prism extends from:

$$-\frac{L}{2} \leq x \leq \frac{L}{2}, \quad -\frac{B}{2} \leq y \leq \frac{B}{2} \quad -\frac{H}{2} \leq z \leq \frac{H}{2} \tag{3}$$

So the integral becomes:

$$\begin{aligned}
I_z^{CG} &= \rho_m \int_{-\frac{H}{2}}^{\frac{H}{2}} \int_{-\frac{B}{2}}^{\frac{B}{2}} \int_{-\frac{L}{2}}^{\frac{L}{2}} \left(x^2 + y^2\right)\, dx\, dy\, dz \\
&= \rho_m \left(\int_{-\frac{H}{2}}^{\frac{H}{2}} \int_{-\frac{B}{2}}^{\frac{B}{2}} \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2\, dx\, dy\, dz + \int_{-\frac{H}{2}}^{\frac{H}{2}} \int_{-\frac{B}{2}}^{\frac{B}{2}} \int_{-\frac{L}{2}}^{\frac{L}{2}} y^2\, dx\, dy\, dz \right) \\
&= \rho_m \left(\frac{L^3 BH}{12} + \frac{LB^3 H}{12} \right) \\
&= \frac{\rho_m LBH}{12} \left(L^2 + B^2\right) \\
&= \frac{m}{12} \left(L^2 + B^2\right) \\
&= \frac{17.0677 \cdot 10^6}{12} \left(161^2 + 21.8^2\right) \\
&= \underline{\underline{3.7544 \cdot 10^{10}}}
\end{aligned} \tag{4}$$

Using the conventional inertia definition we get:

$$I_{xy}^{CG} = -\int_V xy\, dm, \quad I_{yx}^{CG} = -\int_V yx\, dm \tag{5}$$

Since $xy = yx$ we can write:

$$\begin{aligned} I_{xy}^{CG} = \quad I_{yx}^{CG} &= -\int_V yx\, dm \\ &= -\rho_m \int_{-\frac{H}{2}}^{\frac{H}{2}} \int_{-\frac{B}{2}}^{\frac{B}{2}} \int_{-\frac{L}{2}}^{\frac{L}{2}} yx\, dx\, dy\, dz \\ &= -\rho_m \int_{-\frac{H}{2}}^{\frac{H}{2}} \int_{-\frac{B}{2}}^{\frac{B}{2}} y \left[\frac{1}{2}x^2\right]_{x=-\frac{L}{2}}^{x=\frac{L}{2}} dy\, dz \\ &= -\rho_m \int_{-\frac{H}{2}}^{\frac{H}{2}} \int_{-\frac{B}{2}}^{\frac{B}{2}} y \left(\frac{1}{2}\left(\frac{L}{2}\right)^2 - \frac{1}{2}\left(-\frac{L}{2}\right)^2\right) dy\, dz \\ &= \underline{0} \end{aligned} \tag{6}$$

1.2 1b

The parallel-axis theorem states:

$$I_z^{CO} = I_z^{CG} + md^2 \tag{7}$$

Since $r_{bg}^b = \left[x_g = -3.7, y_g = 0, f_g = \frac{H}{2}\right]$ we get:

$$\begin{aligned} d^2 &= x_g^2 + y_g^2 \\ &= 3.7^2 \end{aligned} \tag{8}$$

Inserting d^2 from equation 8 and $I_{z,prism}^{CG}$ from equation 4 the equation becomes:

$$\begin{aligned} I_{z,prism}^{CO} &= \frac{m}{12}\left(L^2 + B^2\right) + m \cdot 3.7^2 \\ &= m\left(\frac{L^2 + B^2}{12} + 3.7^2\right) \\ &= 17.0677 \cdot 10^6 \left(\frac{161^2 + 21.8^2}{12} + 3.7^2\right) \\ &= \underline{\underline{3.7777 \cdot 10^{10}}} \end{aligned} \tag{9}$$

The ratio between the moments of inertia of the prism and real ship is:

$$R = \frac{I_{z,prism}^{CO}}{I_{z,boat}^{CO}} = \frac{3.7777 \cdot 10^{10}}{2.1732 \cdot 10^{10}} = \underline{\underline{1.7383}} \tag{10}$$

1.3 1c

The velocity CG can be expressed in terms of the velocity CO using the rigid-body kinematic relation:

$$\mathbf{v_{CG}} = \mathbf{v_{CO}} + \boldsymbol{\omega} \times \boldsymbol{r_{bg}^b} \tag{11}$$

For the three degrees of freedom we have:

$$\mathbf{v_{CO}} = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix}, \quad \boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ r \end{bmatrix} \tag{12}$$

Using the position vector from CO to CG the cross product becomes:

$$\boldsymbol{\omega} \times \boldsymbol{r_{bg}^b} = \begin{bmatrix} -ry_g \\ rx_g \\ 0 \end{bmatrix} \tag{13}$$

Therefor:

$$\begin{bmatrix} u_{CG} \\ v_{CG} \\ r \end{bmatrix} = \begin{bmatrix} u - ry_g \\ v + rx_g \\ r \end{bmatrix} \tag{14}$$

This can be written in form as:

$$\mathbf{v_{CG}} = \mathbf{H}\left(\boldsymbol{r}_{\boldsymbol{bg}}^{\boldsymbol{b}}\right)\mathbf{v_{CO}} \tag{15}$$

Where the transformation matrix is:

$$\mathbf{H}\left(\boldsymbol{r}_{\boldsymbol{bg}}^{\boldsymbol{b}}\right) = \begin{bmatrix} 1 & 0 & -y_g \\ 0 & 1 & x_g \\ 0 & 0 & 1 \end{bmatrix} \tag{16}$$

If the coordinate origin were located at the center of gravity, the mass matrix would be:

$$\mathbf{M_{RB}^{CG}} = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & I_z^{CG} \end{bmatrix} \tag{17}$$

Using equation 3.29 from [1] we get:

$$\begin{aligned} \mathbf{M_{RB}^{CO}} &= \mathbf{H}^\top\left(\boldsymbol{r}_{\boldsymbol{bg}}^{\boldsymbol{b}}\right)\mathbf{M_{RB}^{CG}}\mathbf{H}\left(\boldsymbol{r}_{\boldsymbol{bg}}^{\boldsymbol{b}}\right) \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -y_g & x_g & 1 \end{bmatrix}\begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & I_z^{CG} \end{bmatrix}\begin{bmatrix} 1 & 0 & -y_g \\ 0 & 1 & x_g \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -y_g & x_g & 1 \end{bmatrix}\begin{bmatrix} m & 0 & -my_g \\ 0 & m & mx_g \\ 0 & 0 & I_z^{CG} \end{bmatrix} \\ &= \begin{bmatrix} m & 0 & -my_g \\ 0 & m & mx_g \\ -my_g & mx_g & I_z^{CG} + m\left(x_g^2 + y_g^2\right) \end{bmatrix} \end{aligned} \tag{18}$$

From equation 7 and equation 8 we know that:

$$I_z^{CO} = I_z^{CG} + m\left(x_g^2 + y_g^2\right) \tag{19}$$

Therefor:

$$\begin{aligned} \mathbf{M_{RB}^{CO}} &= \begin{bmatrix} m & 0 & -my_g \\ 0 & m & mx_g \\ -my_g & mx_g & I_z^{CO} \end{bmatrix} \\ &= \begin{bmatrix} m & 0 & 0 \\ 0 & m & mx_g \\ 0 & mx_g & I_z^{CO} \end{bmatrix} \end{aligned} \tag{20}$$

For planar motion, one valid form of the Coriolis matrix at CG is:

$$\mathbf{C_{RB}^{CG}} = \begin{bmatrix} 0 & 0 & -mv_{CG} \\ 0 & 0 & mu_{CG} \\ mv_{CG} & -mu_{CG} & 0 \end{bmatrix} \tag{21}$$

From equation 14 we have:

$$u_{CG} = u - ry_g, \quad v_{CG} = v + rx_g \tag{22}$$

Substituting these into the Coriolis matrix gives:

$$\mathbf{C_{RB}^{CG}} = \begin{bmatrix} 0 & 0 & -m\left(v + rx_g\right) \\ 0 & 0 & m\left(u - ry_g\right) \\ m\left(v + rx_g\right) & -m\left(u - ry_g\right) & 0 \end{bmatrix} \tag{23}$$

To simplify the multiplication, we define:

$$a = -m(v + rx_g), \quad b = m(u - ry_g) \tag{24}$$

Using equation 3.30 from [1] we get:

$$\begin{aligned}
\mathbf{C}_{\mathbf{RB}}^{\mathbf{CO}} &= \mathbf{H}^\top \left(\boldsymbol{r}_{\boldsymbol{bg}}^{\boldsymbol{b}}\right) \mathbf{C}_{\mathbf{RB}}^{\mathbf{CG}} \mathbf{H} \left(\boldsymbol{r}_{\boldsymbol{bg}}^{\boldsymbol{b}}\right) \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -y_g & x_g & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & a \\ 0 & 0 & b \\ -a & -b & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -y_g \\ 0 & 1 & x_g \\ 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -y_g & x_g & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & a \\ 0 & 0 & b \\ -a & -b & ay_g - bx_g \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & a \\ 0 & 0 & b \\ -a & -b & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & -m\left(v + rx_g\right) \\ 0 & 0 & m\left(u - ry_g\right) \\ m\left(v + rx_g\right) & -m\left(u - ry_g\right) & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & -m\left(v + rx_g\right) \\ 0 & 0 & mu \\ m\left(v + rx_g\right) & -mu & 0 \end{bmatrix}
\end{aligned} \tag{25}$$

1.4 1d

A matrix is skew-symetric if its transpose is equal to its negativ:

$$A^\top = -A \tag{26}$$

The transpose is:

$$\left(\mathbf{C}_{\mathbf{RB}}^{\mathbf{CO}}\right)^\top = \begin{bmatrix} 0 & 0 & m\left(v + rx_g\right) \\ 0 & 0 & -mu \\ -m\left(v + rx_g\right) & mu & 0 \end{bmatrix} \tag{27}$$

The negative is:

$$-\mathbf{C}_{\mathbf{RB}}^{\mathbf{CO}} = \begin{bmatrix} 0 & 0 & m\left(v + rx_g\right) \\ 0 & 0 & -mu \\ -m\left(v + rx_g\right) & mu & 0 \end{bmatrix} \tag{28}$$

Comparing the two matrixes we obtain:

$$\left(\mathbf{C}_{\mathbf{RB}}^{\mathbf{CO}}\right)^\top = -\mathbf{C}_{\mathbf{RB}}^{\mathbf{CO}} \tag{29}$$

Skew-symmetry is important because it ensures that Coriolis and centripetal forces do not add or remove energy from the system. This means the ship's kinetic energy is conserved when no external forces do work.

1.5 1e

The rigid-body Coriolis matrix can be written in a form that only depends on the yaw rate r, rather than the surge and sway velocities u and v.

For irrational ocean currents, the current velocity is given by

$$\nu_c = \begin{bmatrix} u_c \\ v_c \\ 0 \end{bmatrix} \tag{30}$$

The relative velocity is

$$\nu_r = \nu - \nu_c \tag{31}$$

Since the current has no yaw-rate component, the yaw rate remains the same in both ν and ν_r. Therefor for the yaw–rate-only representation,

$$\mathbf{C}^{\mathbf{CO}}_{\mathbf{RB}}(\nu) = \mathbf{C}^{\mathbf{CO}}_{\mathbf{RB}}(\nu_r) \tag{32}$$

This property is useful because the Coriolis matrix does not change when we use relative velocity instead of absolute velocity. It makes it easier to include ocean currents in the ship's equations of motion.

2 Problem 2

2.1 2a

The displaced water volume ∇ for a floating vessel can be fund by using the following equation from Fossen[1]:

$$mg = \rho g \nabla \tag{33}$$

$$\begin{aligned}
\nabla &= \frac{m}{\rho} \\
\nabla &= \frac{17.0677 \cdot 10^6}{1025} \\
\nabla &= 1.6651 \cdot 10^4
\end{aligned}$$

However, for a regular rectangular prism , we can use the geometry directly to calculate displacement:

$$\begin{aligned}
\nabla &= LBT \\
\nabla &= 161 \cdot 21.8 \cdot 4.47 \\
\nabla &= 1.6636 \cdot 10^4
\end{aligned}$$

2.2 2b

The waterplane area A_{wp} can be calculated by using the equation:

$$A_{wp} = LB. \tag{34}$$

Hence, waterplane area given by $A_{wp} = 3.5098 \cdot 10^3$

Z_{hs} can be fund u using the following equation[1]:

$$Z_{hs} = -\rho \nabla \nabla(z). \tag{35}$$

For small perbutaions, it is common to assume that the waterplane area stays constant for small perbutaions for conventional ships[1]. This means that . Using this assumption, we get $\nabla\nabla(z) \approx A_{wp}z$, and Equation 35 can be rewritten as:

$$Z_{hs} \approx -\rho g A_{wp} z.$$

as describet in Fossen[1]. This resluts in the expression:

$$Z_{hs} \approx -3.5292 \cdot 10^7 z$$

2.3 2c

Using the equation given in the task description, we get:

$$T_3 \approx 2\pi\sqrt{\frac{2T}{g}}$$
$$T_3 \approx 6.1766$$

2.4 2d

The expressions for GM_T and GM_L are given by:

$$GM_T = BM_T - BG \tag{36a}$$
$$GM_L = BM_L - BG \tag{36b}$$

We find BG by using $BG = KG - KB$ from Fossen[1], where $KG = H/2$is how far above the keel line CG is located, and $KB = T/2$ is the length of the draft divided by 2. For small inclinations we can make the approximations

$$BM_T = \frac{I_T}{\nabla}$$
$$BM_L = \frac{I_L}{\nabla}$$

We have a rectangular water plane area, which means that $I_T = \frac{1}{12}B^3L$ and $I_L = \frac{1}{12}BL^3$.

The analytical expressions for GM_T and GM_L are given by:

$$GM_T = \frac{1}{12\nabla}B^3L - BG \tag{37a}$$
$$GM_L = \frac{1}{12\nabla}BL^3 - BG \tag{37b}$$

We use this to calculate the numerical values:

$$GM_T \approx 2.8253$$
$$GM_L \approx 450.1838$$

2.5 2e

According to definition 4.2[1] the vessel will be metacentrically stable if both GM_T and GM_L is larger than 0. This is the case for our system, hence the system is metacentrically stable. Furthermore, the definitions specifies that for the situation :

$$GM_L > GM_{T,min} > 0$$

the system is longitudinal metacentrically stable.

3 Problem 3

3.1 3a

The added mass matrix is reduced to three degrees of freedom by keeping only surge, sway and yaw components, corresponding to rows and columns 1, 2 and 6 of the original matrix. This gives:

$$\mathbf{M_A} = -\begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{r}} \end{bmatrix} \tag{38}$$

Since we assume no coupling between surge and the sway-yaw subsystem:

$$X_{\dot{v}} = X_{\dot{r}} = Y_{\dot{u}} = N_{\dot{u}} = 0 \tag{39}$$

And since the added mass matrix is symmetric we get:

$$\mathbf{M_A} = \mathbf{M_A}^\top \quad \Longrightarrow \quad Y_{\dot{r}} = N_{\dot{v}} \tag{40}$$

Therefor:

$$\begin{aligned} \mathbf{M_A} &= -\begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & Y_{\dot{r}} \\ 0 & Y_{\dot{r}} & N_{\dot{r}} \end{bmatrix} \\ &= -\begin{bmatrix} 8.9830 \cdot 10^5 & 0 & 0 \\ 0 & 5.1996 \cdot 10^6 & -9.3677 \cdot 10^5 \\ 0 & -9.3677 \cdot 10^5 & 2.4283 \cdot 10^{10} \end{bmatrix} \end{aligned} \tag{41}$$

3.2 3b

We can get $\mathbf{M_A}$ back to 6×6 by setting all other coefficients than surge, sway and yaw to zero:

$$\mathbf{M_A} = -\begin{bmatrix} X_{\dot{u}} & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{\dot{v}} & 0 & 0 & 0 & Y_{\dot{r}} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & N_{\dot{v}} & 0 & 0 & 0 & N_{\dot{r}} \end{bmatrix} \tag{42}$$

Now we have the four blocks:

$$\begin{aligned} A_{11} &= -\begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad A_{12} = -\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & Y_{\dot{r}} \\ 0 & 0 & 0 \end{bmatrix} \\ A_{21} &= -\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & N_{\dot{v}} & 0 \end{bmatrix} \quad A_{22} = -\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & N_{\dot{r}} \end{bmatrix} \end{aligned} \tag{43}$$

For six degrees of freedom the velocity vector is:

$$\nu = \begin{bmatrix} u & v & w & p & q & q \end{bmatrix}^\top \tag{44}$$

Since we only consider surge, sway and yaw we get:

$$\nu_1 = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix}, \quad \nu_2 = \begin{bmatrix} 0 \\ 0 \\ r \end{bmatrix} \tag{45}$$

Then we get:

$$\begin{aligned} A_{11}\nu_1 &= -\begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} = \begin{bmatrix} -X_{\dot{u}}u \\ -Y_{\dot{v}}v \\ 0 \end{bmatrix} \\ A_{12}\nu_1 &= -\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & Y_{\dot{r}} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ r \end{bmatrix} = \begin{bmatrix} 0 \\ -Y_{\dot{r}}r \\ 0 \end{bmatrix} \\ A_{21}\nu_2 &= -\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & N_{\dot{v}} & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -N_{\dot{v}}v \end{bmatrix} \\ A_{22}\nu_2 &= -\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & N_{\dot{r}} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ r \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -N_{\dot{r}}r \end{bmatrix} \end{aligned} \tag{46}$$

Then we can calculate:

$$\begin{aligned} a = A_{11}\nu_1 + A_{12}\nu_1 &= \begin{bmatrix} -X_{\dot{u}}u \\ -Y_{\dot{v}}v - Y_{\dot{r}}r \\ 0 \end{bmatrix} \\ b = A_{21}\nu_2 + A_{22}\nu_2 &= \begin{bmatrix} 0 \\ 0 \\ -N_{\dot{v}}v - N_{\dot{r}}r \end{bmatrix} \end{aligned} \tag{47}$$

The skew matrices then becomes:

$$\begin{aligned} -S(a) &= \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u \\ -Y_{\dot{v}}v - Y_{\dot{r}}r & X_{\dot{u}}u & 0 \end{bmatrix} \\ -S(b) &= \begin{bmatrix} 0 & -N_{\dot{v}}v - N_{\dot{r}}r & 0 \\ N_{\dot{v}}v + N_{\dot{r}}r & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned} \tag{48}$$

Which gives us the Hydrodynamic Coriolis-Centripetal Matrix:

$$C_A(\nu) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & Y_{\dot{v}}v + Y_{\dot{r}}r \\ 0 & 0 & 0 & 0 & 0 & -X_{\dot{u}}u \\ 0 & 0 & 0 & -Y_{\dot{v}}v - Y_{\dot{r}}r & X_{\dot{u}}u & 0 \\ 0 & 0 & Y_{\dot{v}}v + Y_{\dot{r}}r & 0 & -N_{\dot{v}}v - N_{\dot{r}}r & 0 \\ 0 & 0 & -X_{\dot{u}}u & N_{\dot{v}}v + N_{\dot{r}}r & 0 & 0 \\ -Y_{\dot{v}}v - Y_{\dot{r}}r & X_{\dot{u}}u & 0 & 0 & 0 & 0 \end{bmatrix} \tag{49}$$

Since we only care about surge, sway and yaw we keep rows and columns 1, 2 and 6:

$$C_A(\nu) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u \\ -Y_{\dot{v}}v - Y_{\dot{r}}r & X_{\dot{u}}u & 0 \end{bmatrix} \tag{50}$$

4 Problem 4

4.1 4a

In figure 1 shows the result of the simulation with the hydrodynamic added mass effects.

4.2 4b

In task 4b diagonal linear damping, nonlinear surge damping and cross-flow surge damping was added into the system. The result of this addition can be seen in figure 2.

4.3 4c

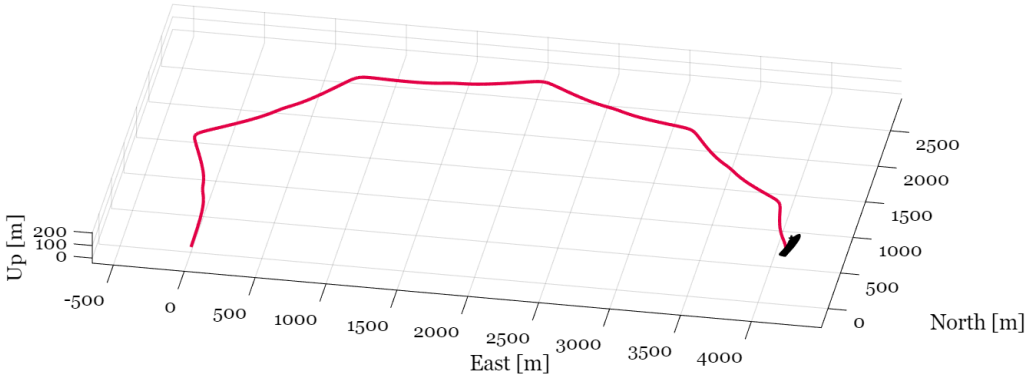
Figure 3 shows the systems response to a step change in rudder position, while figure 4 shows the response to a step change in the propeller speed.

4.4 4d

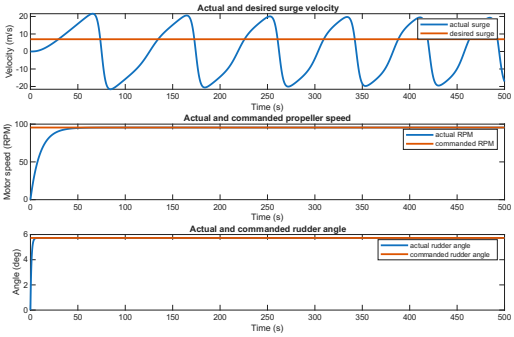
The flow chart depicted in 5 is drawn in Simulink, and shows a visual representation of how equation

$$M_{RB}\dot{\nu} + C_{RB}\nu = \tau_{RB} \tag{51}$$

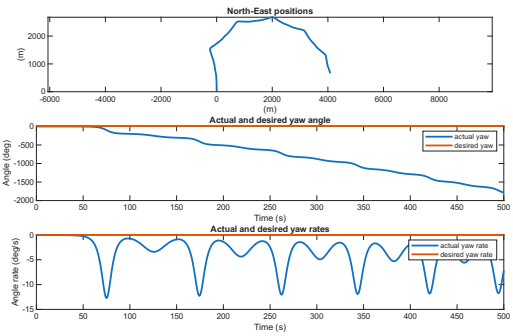
can be used in a simulation of a vessels dynamics. The diagram also illustrates the entire state-equation, and how the vessel state are propagated over time. The states not highlighted in this task are outlined in pseudocode.



(a) 3-D visualization of trajectory



(b) Actual and desired surge velocity, propeller speed and rudder angle

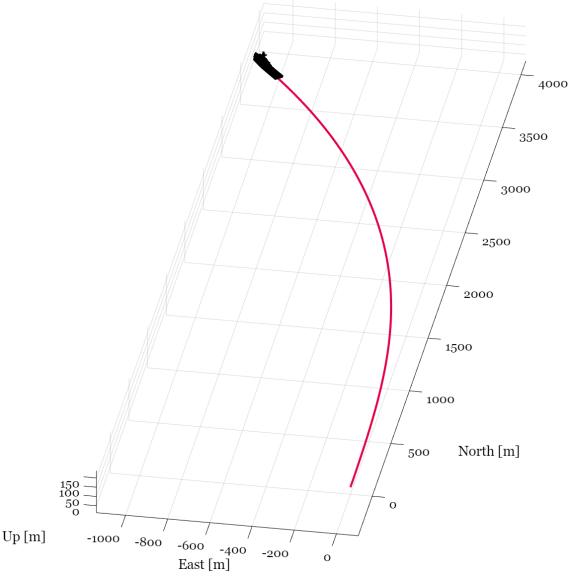


(c) North-East position and actual and desired yaw angle and yaw rate

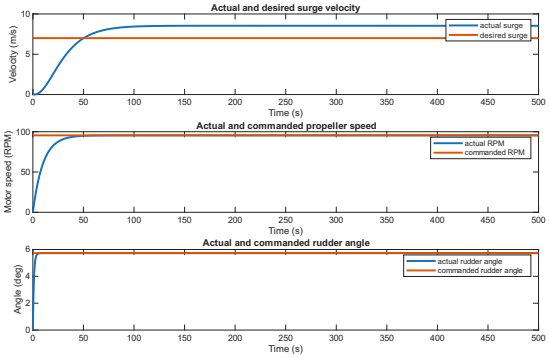
Figure 1: Simulation result of task 4a

References

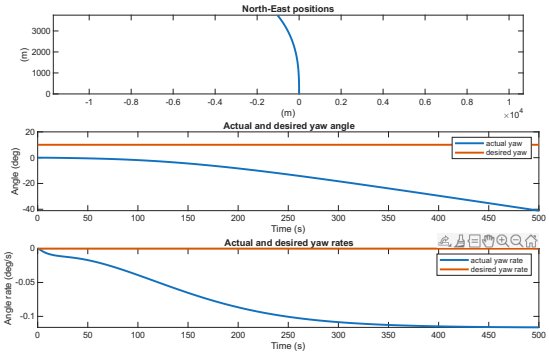
[1] T. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. John Wiley & Sons, 2011.



(a) 3-D visualization of trajectory

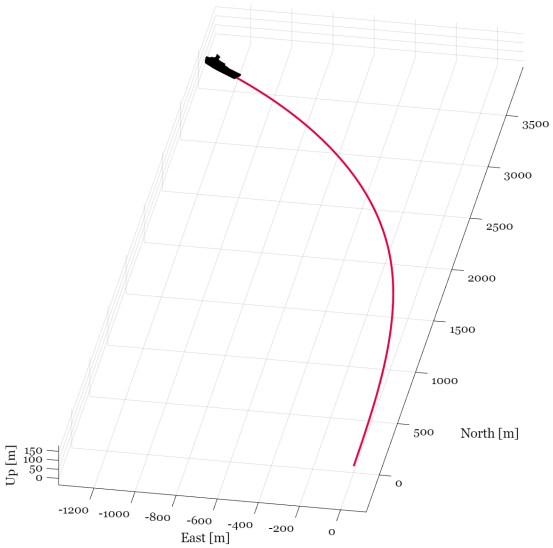


(b) Actual and desired surge velocity, propeller speed and rudder angle

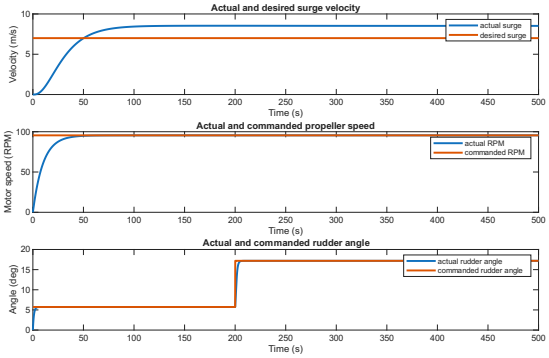


(c) North-East position and actual and desired yaw angle and yaw rate

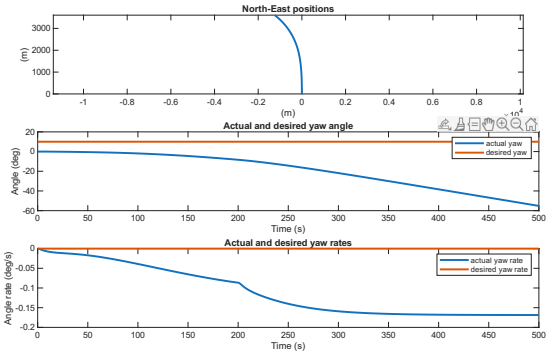
Figure 2: Simulation result of task 4c



(a) 3-D visualization of trajectory

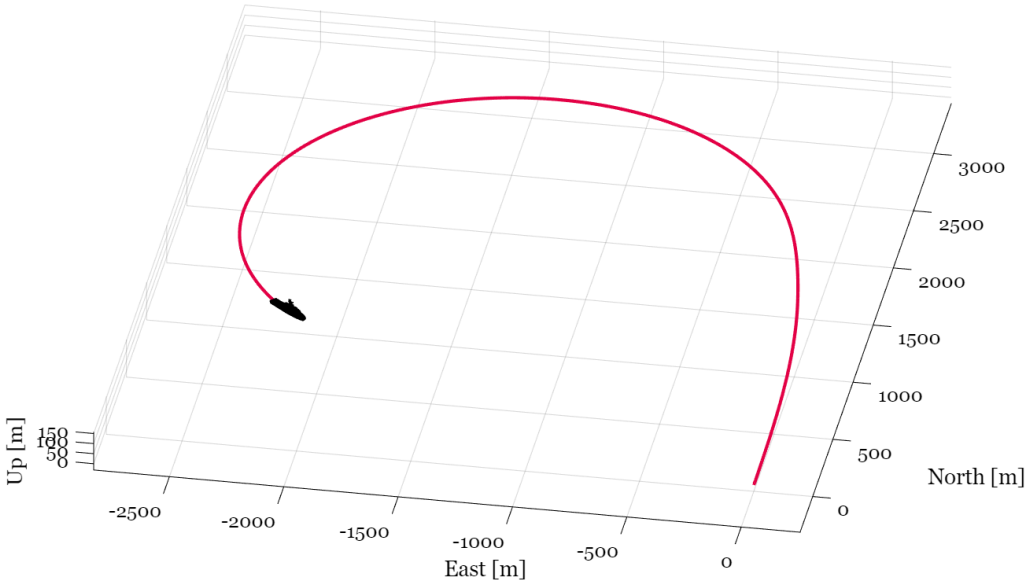


(b) Actual and desired surge velocity, propeller speed and rudder angle

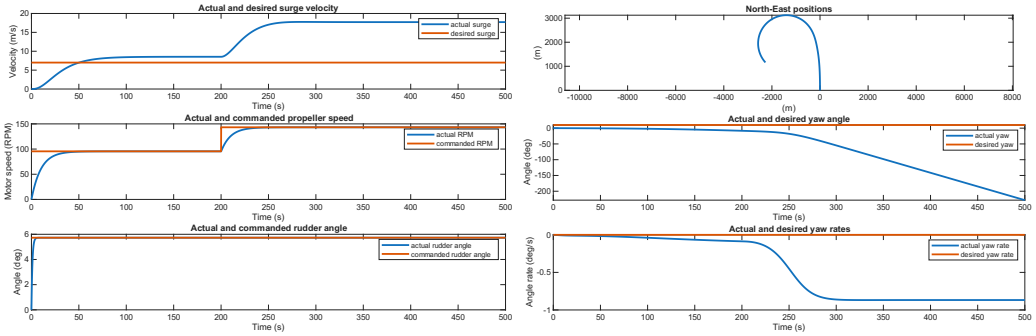


(c) North-East position and actual and desired yaw angle and yaw rate

Figure 3: Simulation result of 4c for the first step response



(a) 3-D visualization of trajectory



(b) Actual and desired surge velocity, propeller speed and rudder angle

(c) North-East position and actual and desired yaw angle and yaw rate

Figure 4: Simulation result of 4c for the second step response

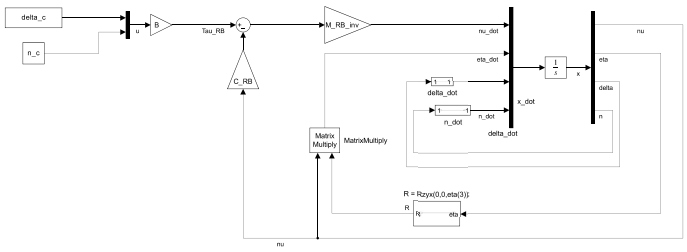


Figure 5: Flowchart drawn in Simulink to illustrate how equation 51 can be used to simulate vessel dynamics.